



Association for Mercy (ASMI)

Born from a profound sense of purpose in **December 2014**, the Association for Mercy officially unfurled its banner and became a legally recognized entity on **May 22, 2018**. This landmark declaration, documented under **N°49/RDA/F.34/SAAJP**, solidified ASMI's commitment to creating lasting positive change.

At the heart of every endeavour, every helping hand extended, and every life touched, lie our unwavering guiding principles:

Compassion • Assistance • Integrity

These are not just words; they are the pillars upon which ASMI builds a brighter, more equitable future.

Association pour la Miséricorde (ASMI)

Née d'un profond sens du devoir en **décembre 2014**, l'Association pour la Miséricorde a officiellement déployé sa bannière et est devenue une entité légalement reconnue le **22 mai 2018**. Cette déclaration marquante, documentée sous le **N°49/RDA/F.34/SAAJP**, a concrétisé l'engagement de l'ASMI à créer un changement positif durable.

Au cœur de chaque initiative, de chaque main tendue et de chaque vie touchée, se trouvent nos principes directeurs inébranlables :

Compassion • Assistance • Intégrité

Ce ne sont pas de simples mots ; ce sont les piliers sur lesquels l'ASMI bâtit un avenir plus lumineux et plus équitable.

CONCEPT NOTE

VisionPlayAI: AI-Powered Spatial Guidance System for Blind Football

1. PROJECT TITLE

VisionPlayAI – AI-Powered Spatial Guidance System for Inclusive Blind Football

2. Background And Problem Statement

Football for visually impaired athletes is an important sport promoting **inclusion, physical well-being, and social participation**. However, blind football players still face significant challenges during training and competition.

Current assistive systems rely mostly on **audible balls, verbal guidance from coaches, or static sound cues**. While helpful, these solutions often provide **limited spatial awareness**, making navigation, ball tracking, and team coordination difficult.

As a result:

- Players may experience **collisions or injuries**
- Training requires **constant external guidance**
- Player autonomy during gameplay remains limited
- Skill development may be slower compared to sighted athletes

Advances in **Artificial Intelligence, real-time localization technologies, and wearable devices** offer an opportunity to dramatically improve the sporting experience for visually impaired athletes.

3. Proposed Solution

VisionPlayAI proposes the development of an **AI-powered wearable assistance kit** designed to enhance spatial awareness for blind football players.

The system will combine:

- **Ultra-Wideband (UWB) localization sensors** to detect the position of players and the ball with high precision
- **AI edge processing** to analyse movement and generate guidance instructions in real time
- **Bone conduction audio devices** delivering directional sound cues without blocking natural hearing

Through this technology, players will receive **real-time spatial feedback** indicating:

- Direction of the ball
- Proximity of teammates and opponents
- Safe navigation paths
- Tactical movement guidance

This creates a **dynamic, intuitive, and safe sporting environment** that significantly improves the independence and performance of visually impaired athletes.

4. Project Objectives

The project aims to:

1. Design and develop a **prototype AI wearable guidance system** for blind football players.
2. Improve **player safety and spatial awareness** during training and competition.
3. Enable **greater autonomy and tactical understanding** for visually impaired athletes.
4. Demonstrate how **AI technologies can advance inclusive sports and accessibility innovation**.

5. Proposed Activities

The project will include the following phases:

Phase 1 – Research and System Design

- Technical architecture design
- Hardware selection
- AI model design for spatial guidance

Phase 2 – Prototype Development

- Assembly of the wearable guidance kit
- Integration of sensors, processing unit, and audio feedback system
- Development of AI software for real-time analysis

Phase 3 – Field Testing

- Pilot tests with two blind football teams
- Evaluation of usability, safety, and performance improvements
- Iteration and refinement of the system

6. Expected Impact

VisionPlayAI is expected to:

- Improve **safety in blind football training and matches**
- Increase **independence and confidence** of visually impaired athletes
- Accelerate **skill development and tactical learning**
- Contribute to **inclusive sports innovation globally**

The technology could also later be adapted for:

- Navigation assistance for visually impaired individuals
- Smart mobility systems
- Inclusive training technologies in other sports.

7. Partnership Request

To successfully develop and test this innovation, we seek collaboration with organizations involved in:

- sports development
- disability inclusion
- technology innovation
- assistive technologies

Potential forms of support may include:

- **Letter of support** for grant and competition applications
- **Access to blind football teams for pilot testing**
- **Technical or advisory collaboration**
- **Visibility and dissemination support**

Such collaboration will strengthen the project's **practical implementation and impact**.

8. Project Lead

Project: VisionPlayAI

Lead Organization :

<<Association pour la Miséricorde (ASMI) >>

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“VisionPlayAI aims to demonstrate how artificial intelligence can empower visually impaired athletes and contribute to building a truly inclusive sporting ecosystem.”